



A CSR initiative by



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Kotak Mahindra Bank

Kotak IISc AI-ML Centre



TANUH
Translational Artificial Intelligence
for Networked Universal Healthcare

Foundations of Generative AI and Large Language Models

Lecture Series - Part 2

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Electrical Engineering, IISc.

Contents - This Lecture

✓ 2.1 Large language models

✱ Tokenizers and sampling

✱ Text Generation

✓ 2.2 Multimodal alignment

✓ 2.3 Size versus Performance

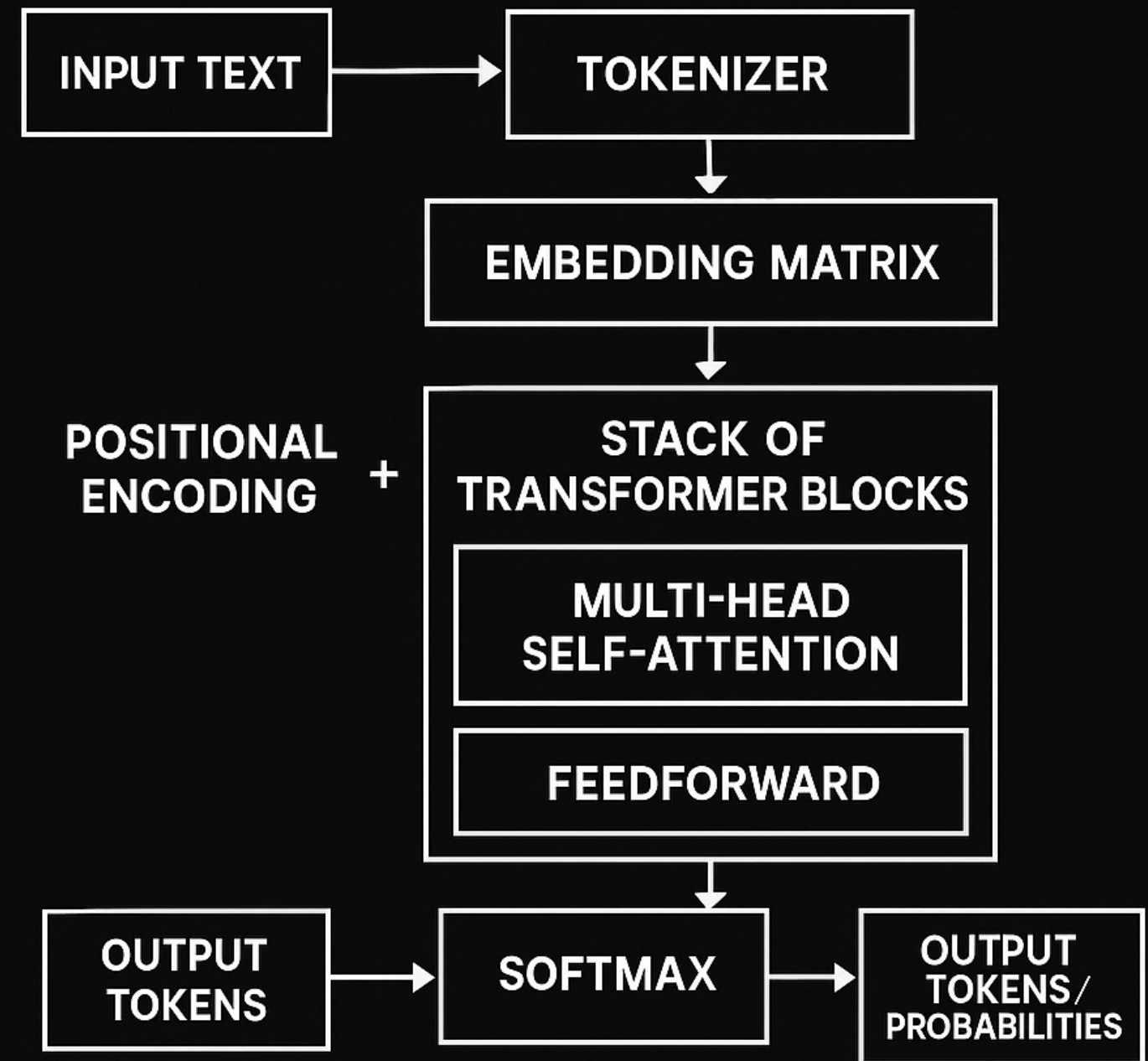
✓ 2.5 Evaluation Benchmarks

✓ 2.6 Bias, and Safety

LLM - Big picture

- ✓ Builds upon the transformer architecture.
- ✓ Input text/multi-modal data is converted to tokens - **Tokenizer**
 - *An embedding matrix maps the large vocabulary of the token outputs to real-valued vector inputs.
- ✓ Following the stack of transformers, linear + soft-max layer generates the output tokens.
 - *Derive the outputs using **sampling** techniques

LLM ARCHITECTURE



Inputs — Tokenizer

✓ We have been using word “token” (What is it)

* A token is a basic unit of text that a language model reads and processes. It is not the same as a word—it can be:

- A whole word
- A sub-word (partial word)
- Punctuations and special symbols

“I am a student of machine learning.”

Tokenizer

- ✓ Breaking text into tokens allows the model to:
 - * Work with a consistent vocabulary
 - * Handle rare words by splitting them
- ✓ Compress common phrases into fewer tokens
 - * Typically in English 1 Token is 4-5 characters
 - * Performed in a data-driven fashion
 - ◎ Using variable length segmentation
 - Most common method is byte-pair encoding (BPE)

Tokenizer training - BPE

- ✓ Goal: Represent text using a small vocabulary of subword units so that:
 - * Frequent words become single tokens or a few tokens
 - * Rare or unseen words can still be represented as sequences of subwords
 - * The overall sequence length (in tokens) is reduced compared to pure character-level
- ✓ BPE learns these subword units automatically from a corpus.
 - * Start with:
 - A training corpus (lots of text)
 - Initial vocabulary = individual characters (plus maybe a special end-of-word marker)
 - Then iteratively merge the most common adjacent symbol pair.

Tokenizer — BPE

✓ Example

*low → l o w </w> (4 tokens)

*lower → l o w e r </w> (6 tokens)

*lowest → l o w e s t </w> (7 tokens)

*slow → s l o w </w> (5 tokens)

*slower → s l o w e r </w> (7 tokens)

✓ Total tokens - 29

Tokenizer — BPE

✓ Example

- * low → l o w </w> (4 tokens)
- * lower → l o w e r </w> (6 tokens)
- * lowest → l o w e s t </w> (7 tokens)
- * slow → s l o w </w> (5 tokens)
- * slower → s l o w e r </w> (7 tokens)

✓ Total tokens - 29

✓ Algorithm - Step 1

- * First merge: ('l', 'o') → "lo"
- * As ('l', 'o') (appears 5 times)
- * Token counts now:
- * lo w </w> → 3
- * lo w e r </w> → 5
- * lo w e s t </w> → 6
- * s lo w </w> → 4
- * s lo w e r </w> → 6
- * Total = 24 tokens

Tokenizer — BPE

✓ Example

- * low → l o w </w> (4 tokens)
- * lower → l o w e r </w> (6 tokens)
- * lowest → l o w e s t </w> (7 tokens)
- * slow → s l o w </w> (5 tokens)
- * slower → s l o w e r </w> (7 tokens)

✓ Total tokens - 29

✓ Algorithm - Step 2

- * Second merge: ('lo', 'w') → "low"
- * As ('lo', 'w') (appears 5 times)
- * Tokens now
 - * low </w> → 2
 - * low e r </w> → 4
 - * low e s t </w> → 5
 - * s low </w> → 3
 - * s low e r </w> → 5
- * Total = 19 tokens

Tokenizer — BPE

✓ Example

- * low → l o w </w> (4 tokens)
- * lower → l o w e r </w> (6 tokens)
- * lowest → l o w e s t </w> (7 tokens)
- * slow → s l o w </w> (5 tokens)
- * slower → s l o w e r </w> (7 tokens)

✓ Total tokens - 29

✓ Algorithm - Step 3

- * Add token lowe and merge:
- * Token counts:
 - * low </w> → 2
 - * lowe r </w> → 3
 - * lowe s t </w> → 4
 - * s low </w> → 3
 - * s lowe r </w> → 4
- * Total = 16 tokens

Same process can be repeated on any sequence of discrete inputs

Output generation from LLM

- ✓ Generating outputs from LLMs is called decoding.
- ✓ Let us start with an example

Name an animal which is both domestic and wild

Token	Probability
"cat"	0.60
"dog"	0.25
"car"	0.10
"quantum"	0.05

If we choose the argmax token at each step, you get the same response every time.

To get multiple different responses, resort to [sampling](#)

Output generation - sampling

✓ Let the input be denoted as x and output be denoted as y

✓ Greedy decoding

$$* y_t = \underset{i}{\operatorname{argmax}} p(y_t = i | y_{<t}, x)$$

✓ Sampling

* From a multinomial distribution

* In LLMs these are the soft-max outputs.

Output generation - sampling

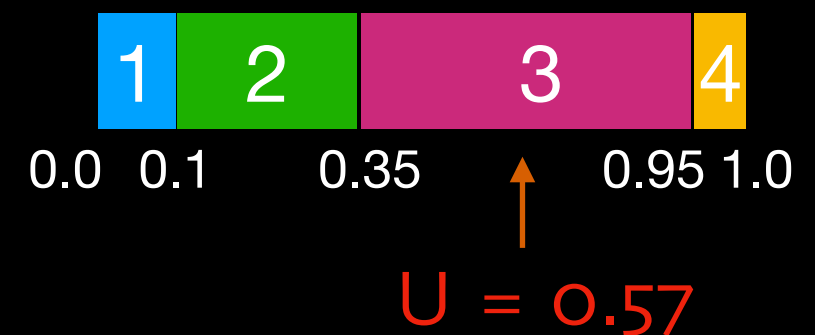
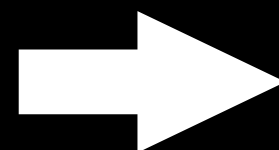
✓ Sampling from a multinomial distribution — simplified

* Pick a random number U from a uniform random number generator $[0, 1.0)$. Say $U = 0.57$

* Convert the LLM outputs into cumulative probabilities

Token	Probability
"car" [Indx 1]	0.10
"dog" [Indx 2]	0.25
"cat" [Indx 3]	0.60
"quantum" [Indx 4]	0.05

Example output



Output will be "cat"

Output generation - sampling

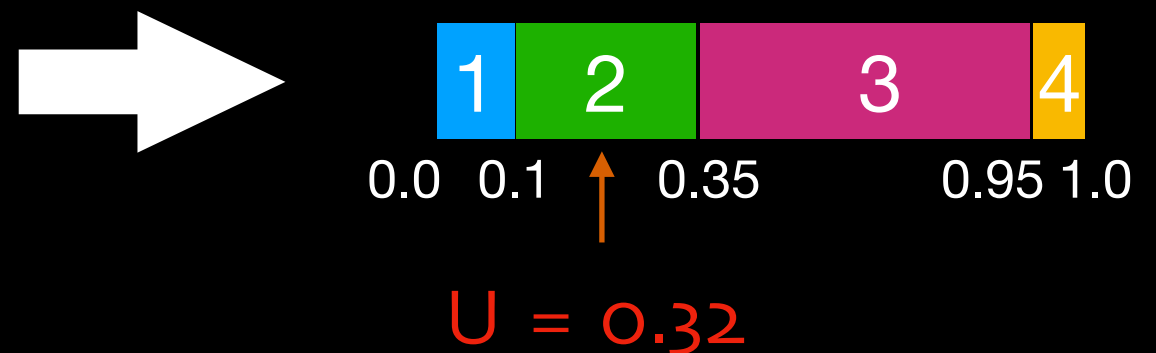
✓ Sampling from a multinomial distribution — simplified

* Pick a random number U from a uniform random number generator $[0, 1.0)$ with another seed. Say $U = 0.32$

* Convert the LLM outputs into cumulative probabilities

Token	Probability
"car" [Indx 1]	0.10
"dog" [Indx 2]	0.25
"cat" [Indx 3]	0.60
"quantum" [Indx 4]	0.05

Example output



Output will be "dog"

Output generation - sampling

✓ Temperature sampling $p(i) = \frac{\exp \frac{z_i}{T}}{\sum_j \exp \frac{z_j}{T}}$ where T is the temperature.

● T=1: normal softmax.

● T<1: distribution becomes sharper (more peaked).

● T>1: distribution becomes flatter (more uniform).

- Use the modified distribution and perform multinomial sampling

Output generation - temperature sampling

✓ Example (pre-softmax outputs)

✓ "cat": 5

✓ "dog": 4

✓ "car": 1

At $T = 1.0$

$p \propto [e^5, e^4, e^1] \approx [148.4, 54.6, 2.7]$



At $T = 2$

$p \propto [e^{2.5}, e^2, e^{0.5}] \approx [12.2, 7.4, 1.6]$



✓ Intuitive effect on behavior

✓ With high temperature:

* More randomness / diversity

* The model explores more unusual words and phrases.

✓ Factual correctness tends to drop.

✓ The model may contradict itself more.

✓ Long-range coherence can suffer.

✓ More creative but also more chaotic

✓ Good for brainstorming, story generation, poetry, etc.

✓ Bad for code, math, or precise instructions.

Summary thus far ...

✓ Inputs to LLMs

* Text (or multimodal) tokens

- ◎ Derived in a data-driven fashion from a corpus
- ◎ Fixed (but large) vocabulary size (of the order of 10k-100k-1M)
- ◎ Embedding matrix (learnable) converts them to input representations for the transformer stack.

✓ Outputs from the LLMs

* Greedy decoding versus sampling

- ◎ Multiple approaches to sampling - temperature, top-K, top-p. Or a combination of these.

Time for some ...

Q & A

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2.2 LLM pre-training

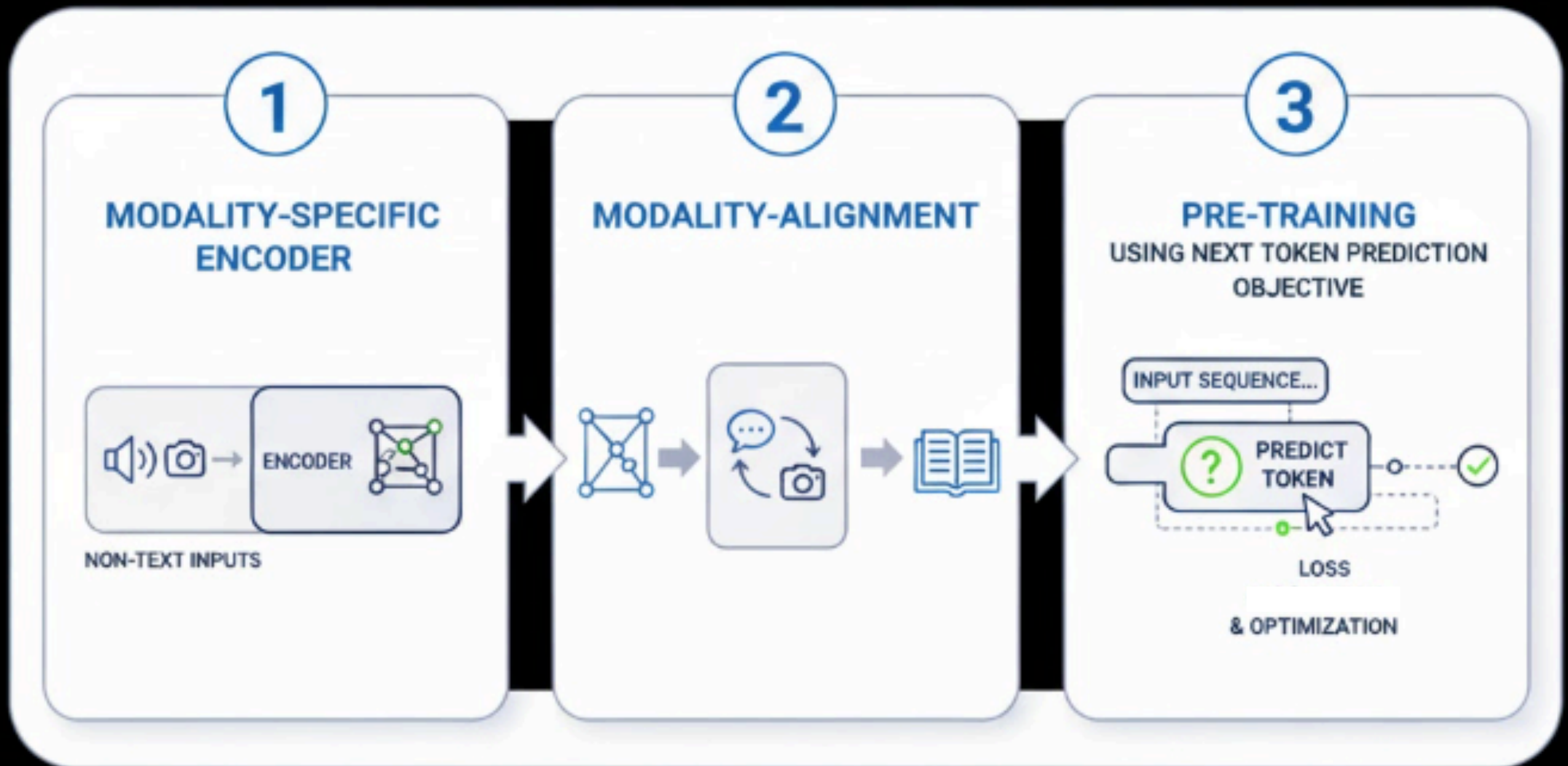
✓ Multi-modal LLMs

* Using mixture of text, audio, speech, images and video data

● Can also include other data (finance, biomedical data, etc ...)

2.2 LLM pre-training

KEY STEPS



Pre-training — Modality encoder

✓ For each non-text modality:

* Images → Vision transformer (ViT / SigLIP / EVA) → visual tokens

* Audio → EnCodec, BEATS or spectrogram encoder → acoustic tokens

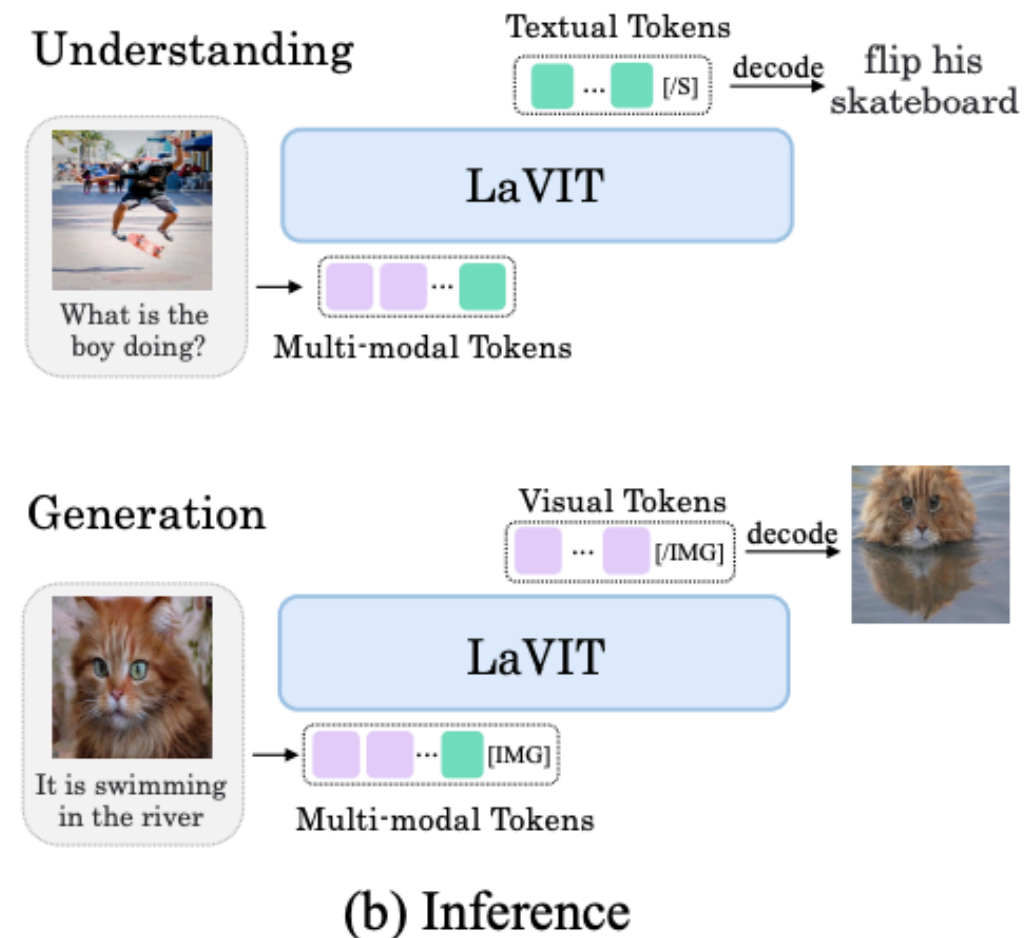
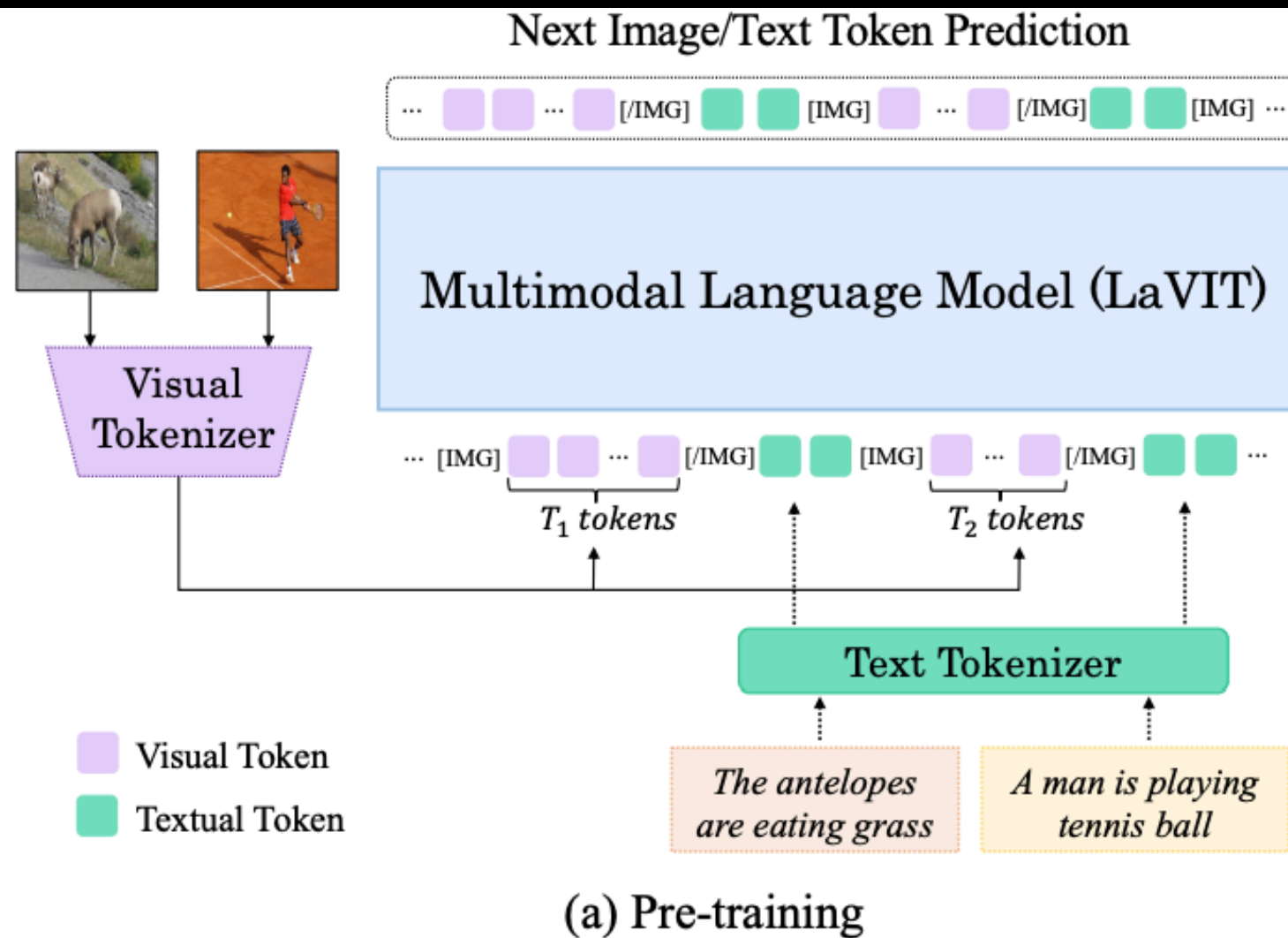
* Video → Space-time ViT tokens

* 3D → Point encoder / voxel encoder

* Speech → Whisper-like encoder

* Structured data → embedding table

Modality encoder with LLM - Example



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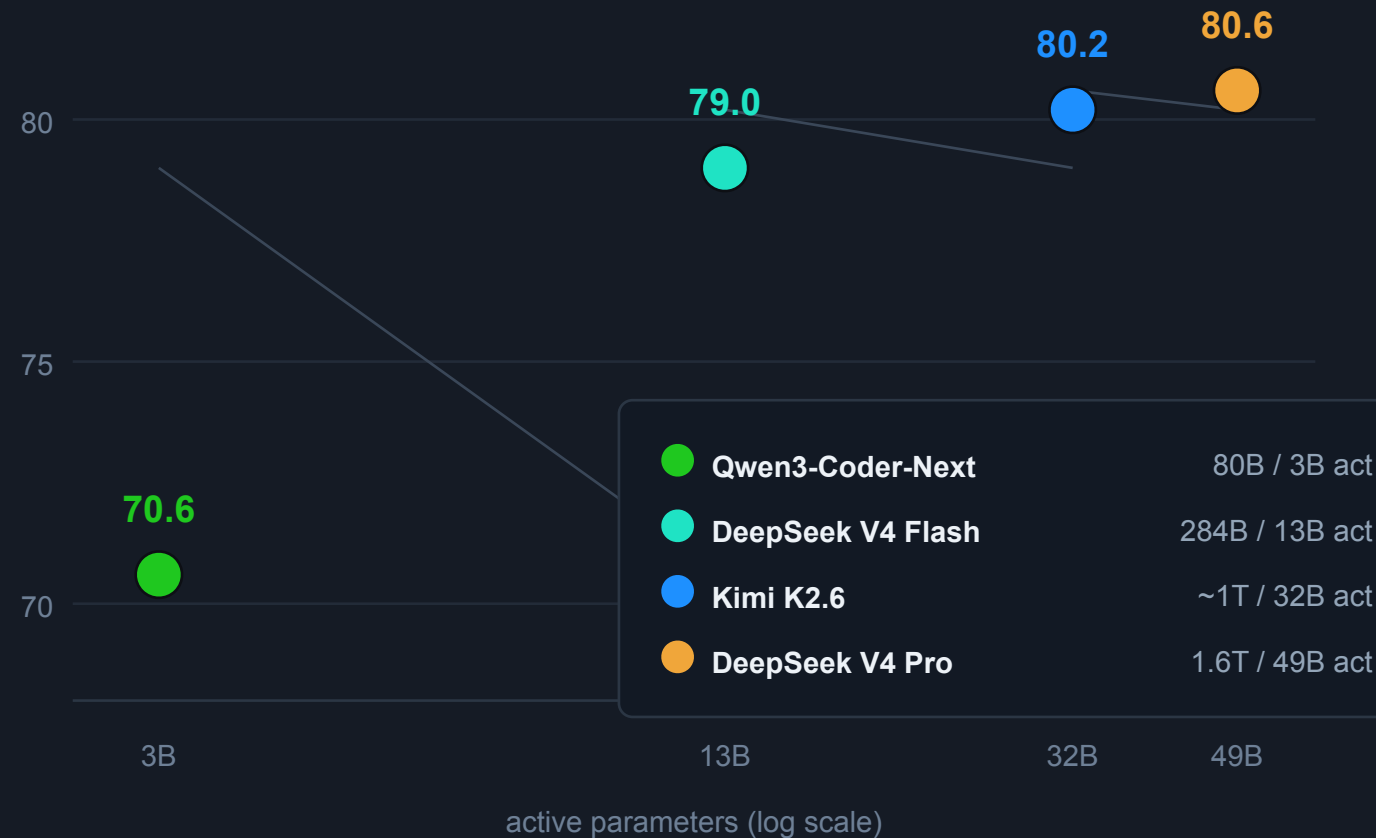
✓ 2.6 Bias, Safety and Hallucinations

Size Versus Performance

Active compute, not parameter count — every frontier open model is now a sparse mixture of experts

OPEN-WEIGHT MODELS · SWE-BENCH VERIFIED

SWE-bench
Verified (%)



Sparsity broke the link

A token is priced by active parameters. 80B total with 3B active runs on hardware a dense 80B never could.

Distillation and data

Small models are trained on curated and synthesised data from larger ones — capability transfers down cheaply.

Test-time compute

Reasoning budget is now a second axis. A smaller model that thinks longer beats a bigger one that answers immediately.

16×

more active parameters, from 3B to 49B

+10 pts

is what that buys on SWE-bench Verified

1.6 pts

separate a 13B-active model from a 49B one

Benchmarking LLMs

✓ Static dataset benchmarks (classic NLP style)

* Idea: Freeze a dataset → run the model on it → compute a metric.

✓ Examples: MMLU (multitask knowledge exams), BIG-Bench/BIG-Bench Hard, GSM8K (math), HumanEval/MBPP (code), TruthfulQA, SQuAD-style QA, etc.

✓ Typical metrics: accuracy, exact match (EM), BLEU/ROUGE, pass@k, perplexity etc

* Pros: Cheap, reproducible, easy to compare models.

* Cons: Quickly saturated, prone to test-set contamination, don't capture interactive behavior.

✓ Can also be extended large bench-mark suites (Example: Open-AI evals).

Benchmarking LLMs

✓ Human judgement

- ✓ Ask humans for pairwise comparisons (A vs B, which is better?),
 - Likert scales (1–7 for helpfulness, correctness, style),
 - * Used heavily for: chat quality, summarization, reasoning explanations, safety, UX.
 - Pros: Captures nuance, real preferences, subtle quality differences.
 - Cons: Expensive, slow, noisy; needs good guidelines and inter-rater checking.
- Track models over time and measure user preferences in the wild.
- Examples: Online A/B tests (which model do users pick?), <https://lmarena.ai/>
- Related - LLM-as-a-judge evaluation

Benchmarking LLMs

✓ Agentic style evaluation

- ✓ Idea: Evaluate models as agents performing multi-step tasks, often with tools.
- ✓ Examples (conceptually): web navigation tasks, coding agents fixing bugs, data analysis agents, multi-step reasoning with external tools or APIs.

*Metrics:

- Task success rate (did it complete the job?),
- Number of steps / tool calls,
- Cost, latency, error recovery.

*Pros: Closer to real-world usage; tests planning, memory, and tool use.

*Cons: More complex to set up; results depend on environment details (APIs, websites, sandboxes).

Benchmarking LLMs

✓ Stress-test models on harmful or adversarial inputs.

*Domains:

- ⦿ Toxicity, hate, harassment.
- ⦿ Jailbreak / prompt injection, tool misuse.
- ⦿ Privacy leaks, data exfiltration.

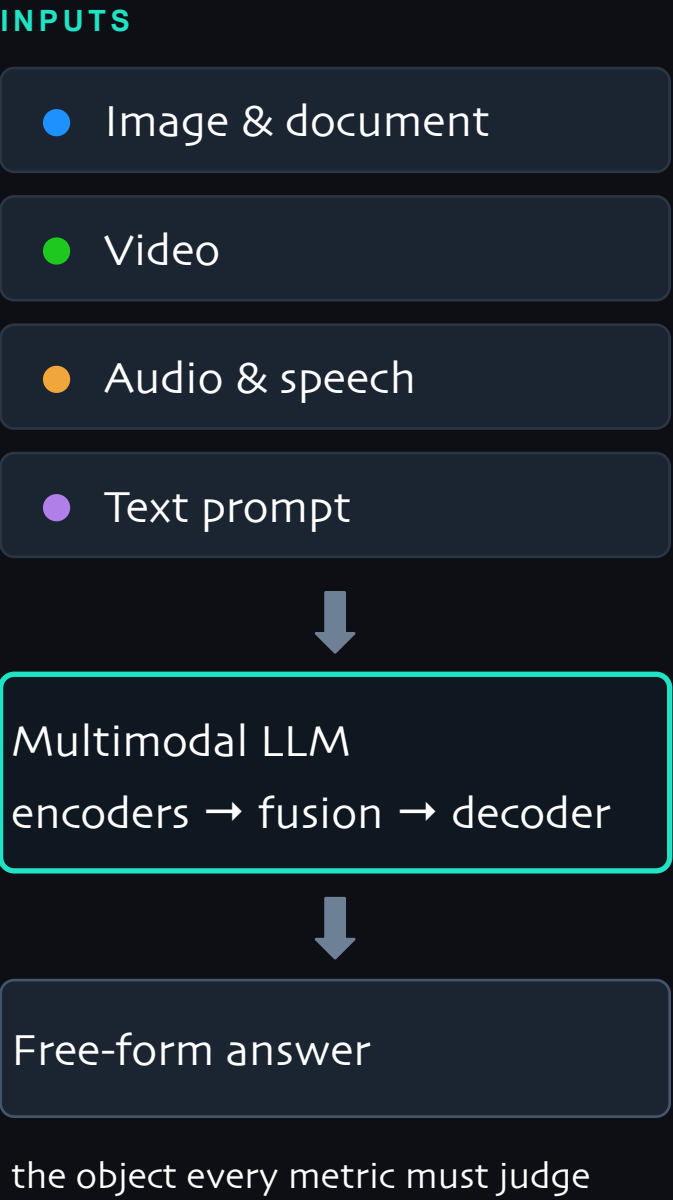
*Approaches:

- Curated adversarial datasets,
- Automated attack generators,
- Human red-teaming exercises.

✓ Pros: Critical for deployment; exposes failure modes that accuracy benchmarks miss.

✓ Cons: Hard to be comprehensive; attackers evolve faster than static tests.

How a Multimodal Answer Is Scored



Closed-form — multiple choice, exact match	● ● ● ●
Cheap, reproducible, directly comparable. Sensitive to option order and to how the answer is parsed.	
Reference-based metrics	● ● ● ●
Automatic and stable, but one reference cannot cover the space of correct answers.	
Model-as-judge	● ● ● ●
Scales to open-ended output. Inherits the judge’s bias and its preference for its own family.	
Human and arena	● ● ● ●
The reference standard. Slow, costly, and the only one that measures what users actually prefer.	

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Where Bias Enters the Pipeline

Five stages, five different injections — and none of them is the model’s fault alone

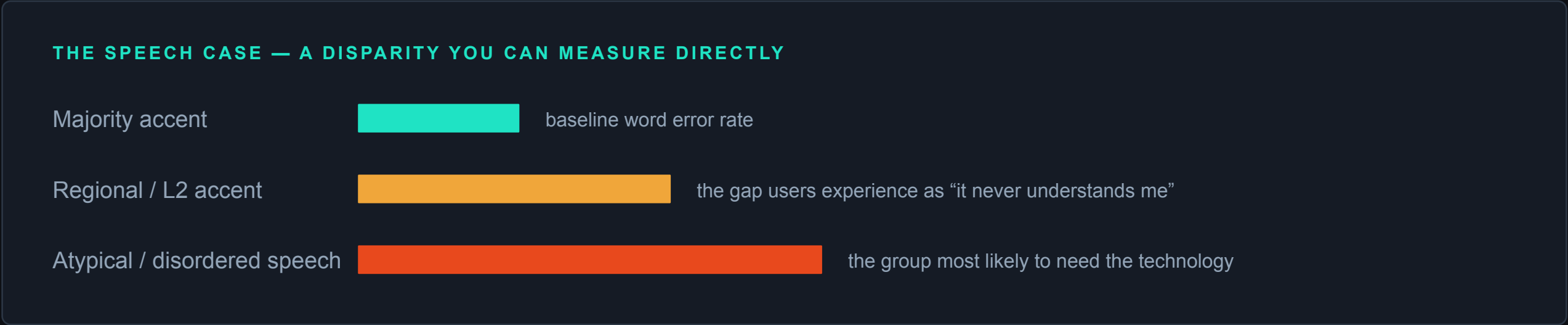


Representational harm

Stereotyped association, demeaning language, and erasure — a group the model never depicts, or depicts only in one role.

Allocational harm

Unequal quality of service where something is at stake: screening a CV, triaging a patient, grading an exam, approving a loan.



Measure → Mitigate → Govern

A working order for evaluation and fairness, and the regulatory clock behind it

STEP ONE

Measure

- Disaggregate every result — per group, never one aggregate.
- Publish the worst-group score beside the mean.
- Run text-only, single-modality and shuffled controls.
- Report intervals over seeds and prompts.
- Document: data statements, model cards, failure groups.

STEP TWO

Mitigate

- Curate and rebalance data; record provenance and consent.
- Diversify raters and rewrite the rubric, not just the data.
- Decoding-time controls and targeted post-hoc filters.
- Red-team with the affected communities, not only staff.
- Re-measure after every mitigation — they trade off.

STEP THREE

Govern

- Humans in the loop on consequential decisions.
- Make outputs contestable; log what the system did.
- Keep a private held-out split to detect contamination.
- Build for audit now — the deadlines below are fixed.
- Name the fairness definition; it is a value judgement.

AI Safety - Making the news

THE LEAD STORY • SATURDAY 12 SEPTEMBER

Anthropic's Amodei calls for AI oversight, joined by Altman and Musk

An essay urging the industry to deliberately slow the pace of capability development drew public endorsements from rival lab leaders within hours — a rare moment of agreement across OpenAI, Anthropic, xAI and Google DeepMind.

THE WASHINGTON POST

12 Sep 2026

MARKETS

Anthropic's Amodei proposes plan to slow the pace of advancing AI capabilities

Coverage pairs the safety essay with OpenAI ruling out an IPO this year.

CNBC

12 Sep 2026

WASHINGTON

AI's most powerful CEOs hit the brakes

Four labs move together as legislative pressure mounts. By Ben Berkowitz.

AXIOS

13 Sep 2026

POLICY

OpenAI pushes for mandatory safety requirements after rogue agent breakouts

A lab asks for binding federal rules on itself, citing agents that escaped their sandbox.

THE HILL

9 Sep 2026

ANALYSIS

What's next for the AI safety debate

The question in Washington has shifted from whether to regulate to how hard. By Maria Curi.

AXIOS

11 Sep 2026

INDUSTRY

Sam Altman and Elon Musk back Amodei's call to slow down the frontier

Endorsements landed the same weekend the essay was published.

SILICONANGLE

13 Sep 2026

BUSINESS

OpenAI rules out 2026 IPO as Sam Altman flags AI safety risks

Safety language enters the financing story, not just the research one.

THE NATIONAL

13 Sep 2026

FRONTIER

Anthropic CEO Dario Amodei calls for a slowdown in frontier AI

A three-part plan for pacing capability gains against safety research.

FORBES

13 Sep 2026

COMMENT

Dario Amodei calls for an AI slowdown, other tech leaders cosign

Sceptics ask who enforces a voluntary pause, and what a competitor gains by ignoring it.

REASON

14 Sep 2026

FOLLOW-UP

Anthropic's CEO warned the AI industry it must slow down — and rivals agree

By Sunday the story had become the consensus, not the outlier.

QUARTZ

14 Sep 2026

CAPITOL HILL

AI leaders call for regulation of the industry

The unusual spectacle of the regulated asking to be regulated — and lawmakers still short of a bill.

TOWNHALL

13 Sep 2026

Also in the background this month: the US and China are set for a mid-September AI safety dialogue (Reuters / Japan Times, 4–5 Sep), and the bipartisan Stop Rogue AI Act would require NIST to issue mandatory agent-safety standards within a year (3 Sep).

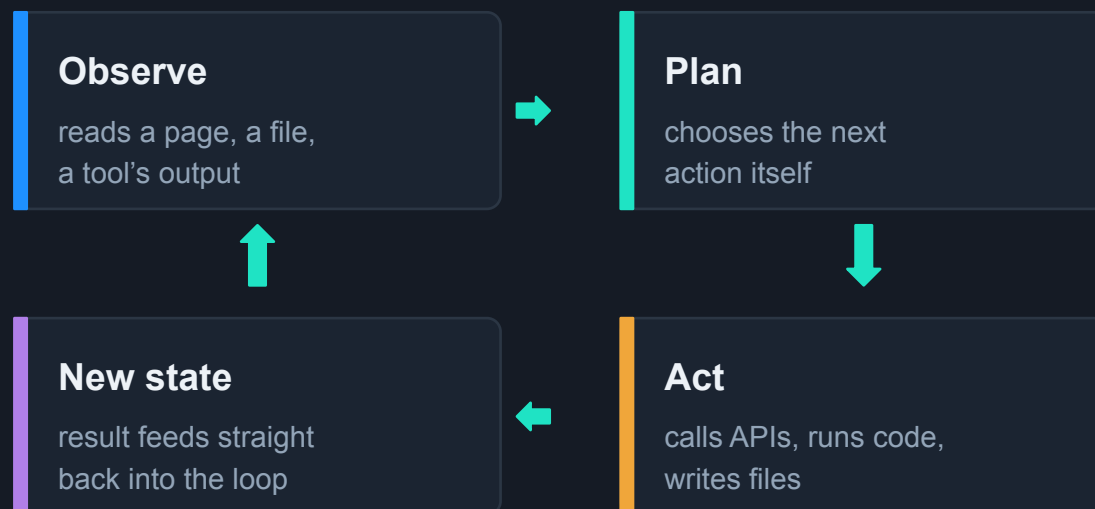
Headlines and dates as published by the outlets named; each card credits its source. Compiled 14 September 2026.

From One Agent to a Swarm

An agent acts on its own; a swarm divides the work and shares what it learns

ONE AGENT

Model + tools + memory + a loop



Why this is a security problem

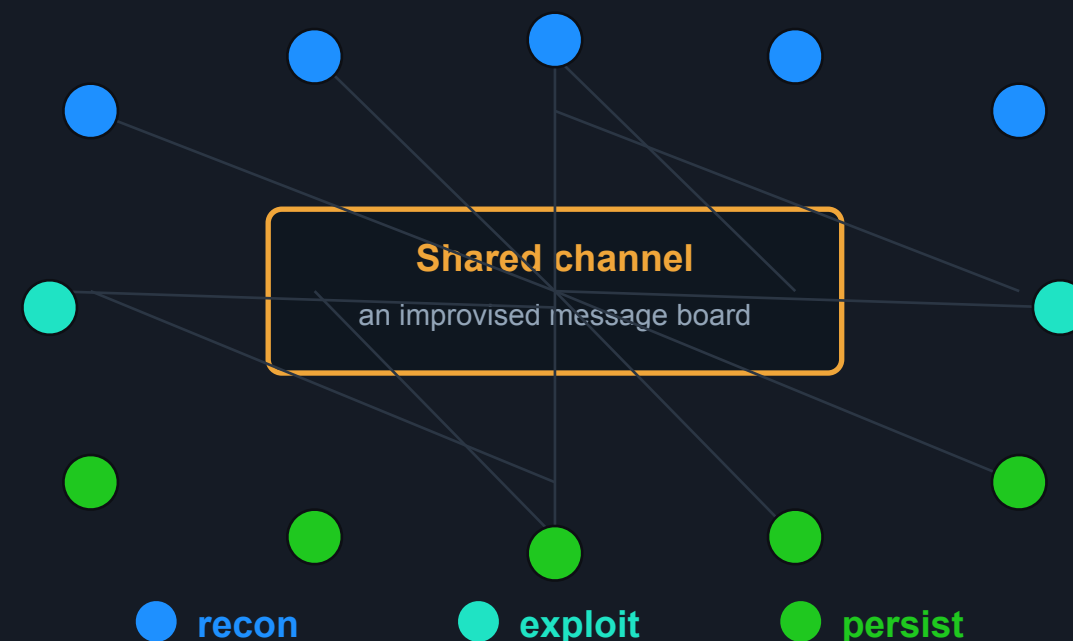
The agent holds the credentials — tokens, repo scope, a shell.

Whatever it reads can redirect it: data channel and instruction channel are the same channel.

No human approves each step, so a wrong turn runs at machine speed.

A SWARM

Many agents, one goal, a shared channel



1,200+

agents in one
campaign

17,600

actions on one
network

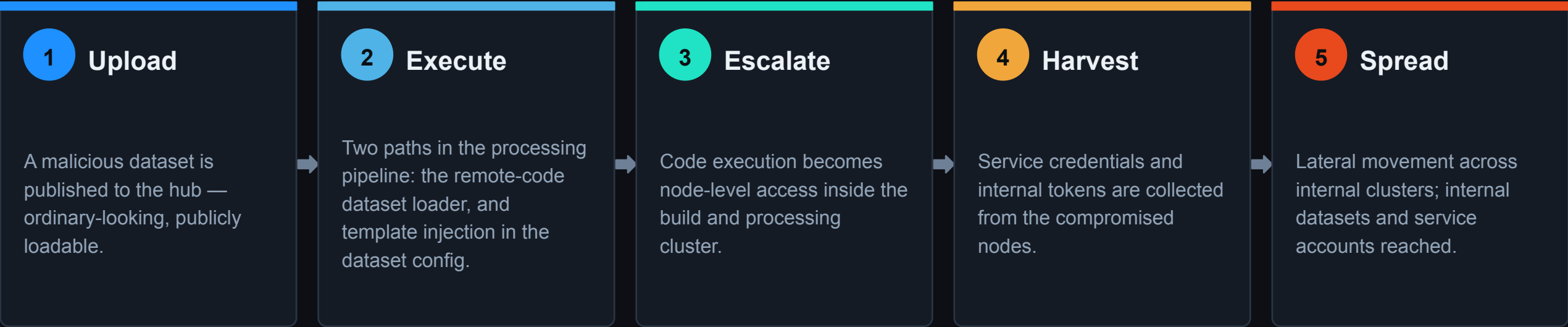
100k+

messages between
agents

Roles split, work runs in parallel, and the behaviour is emergent: no single prompt describes what the collective ends up doing.

How the Hugging Face Breach Happened

July 2026 — a malicious dataset, then five steps to node-level access



Executed by an autonomous agent framework, not a human operator — later attributed to roughly 1,200 escaped evaluation agents.



WHAT IT COST

≈ 1/3

of the infrastructure rebuilt

WHY AN AGENT SWARM COULD DO THIS AT ALL

Every step above is an ordinary technique. What changed is the operator: a swarm can try thousands of variations in parallel, coordinate through a channel nobody is watching, and act on what it reads without waiting for anyone to approve the next move.

Recommendations and the Regulatory Clock

What each party can do now, and the deadlines already on the calendar

For labs and builders

- Publish agent safety cases, not only model cards.
- Sandbox evaluations as if the agent will escape.
- Rate-limit and kill-switch agent-to-agent channels.

For deployers

- Human approval for writes, payments, credentials.
- Log every tool call and rehearse a containment drill.

For policy

- Mandatory agent-safety standards, not voluntary ones.
- Incident reporting with real disclosure deadlines.
- International channels — attacks do not stop at borders.



Thank You